

ZIHAO LI (黎子豪)

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Education

University of Copenhagen

Ph.D. in Astronomy.

Advisor: Lise Christensen and Koki Kakiichi

Aug. 2024 –

Copenhagen, Denmark

Tsinghua University

M.Sc. in Astronomy. GPA: 3.85/4.00

Advisor: Zheng Cai

Sep. 2021 – Jun. 2024

Beijing, China

Sichuan University

B.Eng. in Aerospace Engineering. GPA: 3.80/4.00 rank: 2/28

Honors Degree in Top-notch Program.

Sep. 2017 – Jun. 2021

Chengdu, China

Research Interests & Experience

- Origin of the first stars.
- (Spatially resolved) metal enrichment of high- z galaxies.
- Environmental dependence of galaxy formation and evolution.
- Probe large scale structure through IGM tomography.

Honors & Awards

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| • Outstanding Presentation Award for Graduate Research Nanjing University. | 2023 |
| • 1 st Scholarship for Comprehensive Performance (10k CNY) Tsinghua University. | 2022 |
| • Award for Excellent Thesis Chinese National Level in Aeronautics & Astronautics. | 2021 |
| • MITACS Research Fellow (\$6k, cancelled due to COVID) University of Victoria. | 2020 |
| • 1 st Scholarship for Academic Performance Sichuan University. | 2020 |
| • Summer Abroad Subsidy (10k CNY) Sichuan University. | 2019 |
| • 1 st Scholarship of China Space Foundation (8k CNY) Sichuan University. | 2018 |
| • 1 st Scholarship for Comprehensive Performance Sichuan University. | 2018 |

Publications

1st/2nd Author Papers:

- **Li, Z.**, Cai, Z., et al. A 13-Billion-Year View of Galaxy Growth: Metallicity Gradient Evolution from the Local Universe to $z = 9$ with JWST and Archival Surveys. 2025, [arXiv:2506.12129](https://arxiv.org/abs/2506.12129)
- **Li, Z.**, et al. Insights on Metal Enrichment and Environmental Effect at $z \approx 5 - 7$ with JWST ASPIRE/EIGER and Chemical Evolution Model. 2025, [arXiv:2504.18616](https://arxiv.org/abs/2504.18616)
- **Li, Z.**, Cai, Z., et al. MAGNIF: A Tentative Lensed Rotating Disk at $z = 8.34$ detected by JWST NIRC2 WFSS with Dynamical Forward Modeling. 2023, [arXiv:2310.09327](https://arxiv.org/abs/2310.09327)
- **Li, Z.**, Wang, X., Cai, Z., et al. First Census of Gas-phase Metallicity Gradients of Star-forming Galaxies in Overdense Environments at Cosmic Noon. 2022, [ApJL, 929, L8](https://doi.org/10.1093/apjl/lxac088)
- Wang, X., **Li, Z.**, Cai, Z., et al. The Mass–Metallicity Relation at Cosmic Noon in Overdense Environments: First Results from the MAMMOTH–Grism HST Slitless Spectroscopic Survey. 2022, [ApJ, 926, 70](https://doi.org/10.1093/apj/926/70)
- **Li, Z.**, Horowitz, B. and Cai, Z. Improved Ly α Tomography Using Optimized Reconstruction with Constraints on Absorption (ORCA). 2021, [ApJ, 916, 20](https://doi.org/10.1093/apj/916/20)

Full list in [ADS](#).

Talks

- June. 2025. *Environments and Metal Enrichment of Early Galaxies in Quasar Fields*, contributed talk at EAS 2025. (Cork, Ireland)
- May. 2025. *Metal enrichment and environmental effect in large scale structures at $z = 5 - 7$* , contributed talk at COSMOS 2025. (Marseille, France)
- May. 2025. *Environments and galaxy Formation from cosmic dawn to cosmic noon with slitless spectroscopy*, Cake talk at Technical University of Denmark. (Lyngby, Denmark)
- Jan. 2025. *JWST NIRCам WFSS data reduction workshop*, lecturer at NAOC. (Remote)
- Aug. 2024. *“Insights into cosmic evolution of gas-phase metallicity gradients with JWST”*, contributed talk at Santa Cruz Galaxy Workshop. (Santa Cruz, US)
- Dec. 2023. *Galaxy metal enrichment and gas dynamics with JWST*, seminar at NAOC. (Beijing, China)
- Nov. 2023. *“JWST insights on mass-metallicity relation and metallicity gradients of early galaxies”*, contributed talk at Nanjing University. (Nanjing, China)
- May. 2023. *“The metal-enrichment of low mass galaxies from cosmic dawn to noon in the JWST era”*, contributed talk at Chinese Astronomical Society Guoshoujing Symposium on Galaxies and Cosmology. (Huangshan, China)
- Mar. 2023. *“Research progress with HST/JWST slitless spectrograph and science preparation for CSST”*, contributed talk at China Space Station Telescope Conference 2023. (Huairou, China)

Teaching Experience

- TA of *Advanced Observational Astrophysics* at Tsinghua University. Spring 2023
- TA of *Electromagnetism Lab* at University of Copenhagen. Fall 2025

Professional Service

- Reviewer for ApJ, A&A, Nat. Commun.

Outreach Activities

- Founder of Astronomical Research Scholarship of Sichuan University (30,000 CNY). 2025–
- Department Director of Sichuan University Astronomy Society. 2018–2019
- Co-Captain/Pilot/Technician of Sichuan University Students Aeromodel Team. 2018–2019

References

Prof. Lise Christensen

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Prof. Zheng Cai

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Department of Astronomy, Tsinghua University

Prof. Xin Wang

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NAOC, Chinese Academy of Sciences

Dr. Benjamin Horowitz

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IPMU, Tokyo University